

Dual ordinal spectral ranking of journals and institutions

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Abstract

We develop a dual spectral ranking of journals and institutions that jointly determines prestige scores via a Katz resolvent applied to a block attention matrix constructed from intra-field citation flows. A continuous parameter χ apportions each reference between source and institution channels; a binary block-mask selects among four bibliometrically distinct flow structures, nesting conventional single-mode rankings as special cases. Applied to the economics and commerce literature using OpenAlex data, the method produces stable, interpretable joint rankings whose sensitivity to model parameters is examined systematically.

Keywords: Scientometrics, spectral ranking, dual journal–institution prestige, Katz resolvent, Pinski–Narin

1 Introduction

The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval and exploration. Institutions fund individual researchers, implement training, oversee compliance, establish research organisational units and maintain physical and information infrastructure. Although the status hierarchy that links journals and institutions is a pervasive aspect of the research system (Merton, 1973; McNamee and Willis, 1994; Martin, 1998), bibliometric studies that explore the joint hierarchy of journals and institutions are rare. This paper proposes a joint ordinal spectral ranking of journals and institutions derived from bibliometric data. We hope that the method helps to reveal how status is formed and maintained. We also intend to apply the approach to study the relations between status and research quality assurance.

The introduction of the science citation index (Garfield, 1964) opened a new way to investigate connections between journals and institutions. Citation-based journal ranking in particular became a useful means to manage collections and to select outlets for publications. On the other hand, the misapplication of a journal’s impact factor (JIF) as proxy for the importance of research published in that journal fostered a still-unresolved tension in bibliometric evaluation (Garfield, 1979; MacRoberts and MacRoberts, 2018). Perhaps due to the complexity of disciplines and missions, an Institution Impact Factor has not emerged. Rather, institutional rankings derived from bibliometric indicators of works, authors or journals are often composed with other indicators to form an institutional score. This is the approach in global rankings such as QS, Times Higher Education (THE) and Shanghai Ranking (ARWU), the academic Leiden Ranking and the SCImago institutions rankings, as well as many national performance-based research funding schemes (Zacharewicz et al, 2019).

Prospects for using citation indices to explore research importance and usage measures were considered by (Narin, 1976) who proposed a recursive method for influence weights of journals. Suitably scaled, the journal influence weight is additive Pinski and Narin (1976) within a disciplinary field. Using influence weights for journals in ten science disciplines, Narin added total influence scores for those disciplines in several dozen universities, as the sum of journal-influence-weighted publications. The scores were well-correlated with survey-based rankings, and also with the number of publications, suggesting that survey respondents combined influence and size. Narin’s work built on earlier studies of social networks and developed into the eigenvector approach for journal ranking Bergstrom et al (2008); Vigna (2016); Franceschet (2010). Moving from journal-level to author-level influence weightings, West et al (2013) demonstrated additive ranking of institutions and countries using publications in Social Science. An alternative approach iterates between a publication network linked by the reference lists (without projection) and the corresponding network of authors Ujum et al (2015); Pal and Ruj (2015). The formulation by Cao et al (2023) presents a similar

network, with reference-based article-article links, and the related article-institution links, ranked by a novel and effective iteration over random walks. The authors note that a limiting case of this approach corresponds to the eigenvector (pageRank) method. As noted earlier by Geller (1978), random walks are closely related to the spectral ranking problem discussed in this paper.

Theoretical framing. Following Wouters (1999), a *reference* written by a scientist becomes a *citation*—a directed edge (w_i, w_j) —only after indexer resolution to canonical work records.

We treat citations as structural relations within a functionally differentiated scientific system (Wouters, 1999), setting aside authorial intent. Dual spectral rankings of journals and institutions treat citations as autopoietic communications within science’s communicative system (Social System of Science Citation Theory, SSCT), where individual actors reside in the environment rather than constituting system elements (Tahamtan and Bornmann, 2022). This contrasts with actor-network theory (ANT), which foregrounds heterogeneous agents in assembling citation networks; ANT enters only in the Discussion. The resulting dual rankings are self-referential hierarchies that condition future citations. Our construction extends Breiger (1974): all four blocks of the joint attention matrix are primary flows, not derived projections.

This paper proceeds as follows. In the next section, we construct a framework for computing the attention or influence of journals and institutions coupled by the reference lists of works they publish. We parametrise many assumptions and variations, so we can assess their consequences in our examples. We then construct a corpus of works in the fields of Economics and Business, and present spectral rankings of journals and institutions in these fields alongside sensitivity studies of many parameters. We then discuss the results, including a critique of our methods, and close by mentioning some directions for future work.

2 Journal–institution citation networks

We begin by selecting N_s sources $\mathfrak{S}$ in a field, discipline or category. Then we select N_w works $\mathfrak{W}$ published in $\mathfrak{S}$ within the census window years $t_c = [t_0^c \dots t_1^c]$ or target years $t_t = [t_0^t \dots t_1^t]$. From the reference lists of the set of census works $\mathfrak{W}^c$ we identify the set of target works $\mathfrak{W}^t$ and form the filtered reference set $\mathfrak{R} = \{(i, j) : w_i \in \mathfrak{W}^c, w_j \in \mathfrak{W}^t\}$. Author affiliations in $\mathfrak{W}$ establish a set of authors $\mathfrak{A}'$ and institutions $\mathfrak{U}'$. Let $\bar{w}_u$ be the mean annual work count for institution u . We determine a lower limit τ_U on $\bar{w}_u$ by examining the work-institution frequency distribution and define $\mathfrak{U} \subseteq \mathfrak{U}'$ as the institutions with $\bar{w}_u \geq \tau_U$, with corresponding author set $\mathfrak{A} \subseteq \mathfrak{A}'$. Works that lose all their authorships under this filter are dropped. Each retained work w_i has an *institution weight* under author-fractional counting (Waltman and van Eck, 2015),

$$\omega_{iu} = \sum_{\ell: u \in \Gamma_{i\ell}} \frac{1}{a_i} \frac{1}{u_{i\ell}}, \quad (1)$$

where ℓ indexes the retained authors of work w_i , a_i is their count, and $u_{i\ell}$ the number of retained institutions of author ℓ , so that $\sum_u \omega_{iu} = 1$.

When a work w_i references another work w_j we say that w_i gives attention to w_j . The attention given by a work can be either (a) the number of references, or (b) a fixed amount (Zitt and Small, 2008). Let $R_i = |\{j : (i, j) \in \mathfrak{R}\}|$ be the reference count of work w_i and $\bar{R} = |\mathfrak{R}|/N_w$ the mean reference count over $\mathfrak{W}$. We use the *reference normalisation weight* ρ_i to select case (a) or case (b) respectively:

$$\rho_i = \begin{cases} 1 & \text{(full count)} \\ \bar{R}/R_i & \text{(fixed count)}. \end{cases} \quad (2)$$

We construct the set of *units* [sources or institutions; Pinski and Narin (1976)] $\mathfrak{P} = \mathfrak{S} \cup \mathfrak{U}$ with $N = N_s + N_u$ elements, ordered so that sources occupy indices $1, \dots, N_s$ and institutions indices $N_s + 1, \dots, N$. A reference $(i, j) \in \mathfrak{R}$ gives attention across unit-type pairs: source–source (*SS*), source–institution (*SI*), institution–source (*IS*), and institution–institution (*II*).

For each work w_i we define the source membership one-hot vector $\mathbf{b}_i^S \in \mathbb{R}^{N_s}$ with $[\mathbf{b}_i^S]_s = \mathbf{1}[w_i \in s]$, and the institution weight distribution vector $\mathbf{b}_i^U \in \mathbb{R}^{N_u}$ with $[\mathbf{b}_i^U]_u = \omega_{iu}$ from (1). Attention flow between

pairs of unit types is accumulated as four blocks over $\mathfrak{R}$ as rank-1 outer products,

$$\mathbf{C}_{XY} = \sum_{(i,j) \in \mathfrak{R}} \rho_i \mathbf{b}_i^X (\mathbf{b}_j^Y)^\top, \quad XY \in \{SS, SI, IS, II\}, \quad (3)$$

with $\mathbf{b}_i^X$ weighting the referencing (citing) side and $\mathbf{b}_j^Y$ the referenced (cited) side.

The citation matrix assembles the four blocks:

$$\mathbf{C}(\chi, \mathbf{m}, \rho) = \begin{pmatrix} m_{SS}(1-\chi)^2 \mathbf{C}_{SS} & m_{SI}\chi(1-\chi) \mathbf{C}_{SI} \\ m_{IS}\chi(1-\chi) \mathbf{C}_{IS} & m_{II}\chi^2 \mathbf{C}_{II} \end{pmatrix}, \quad (4)$$

where the binary *block mask* $\mathbf{m} = (m_{SS}, m_{SI}, m_{IS}, m_{II}) \in \{0, 1\}^4$ is bibliometrically meaningful for source-only (1, 0, 0, 0), institution-only (0, 0, 0, 1), bipartite (0, 1, 1, 0), and full-joint (1, 1, 1, 1) combined blocks. The mixing weight χ controls the source–institution balance in the bipartite and full-joint cases and is inert in the single-mode cases. The χ scaling preserves total citation flow across all block combinations, since $(1-\chi)^2 + 2\chi(1-\chi) + \chi^2 = 1$. We could name $\mathbf{C}$ the reference, attention or citation matrix and prefer the latter.

The model parameters are summarised in Table 1. Note that the corpus excludes (a) census work references not giving attention to target works, (b) non-census works giving attention to target works, and (c) authorships that include an omitted institution.

Table 1: Model parameters θ .

Symbol	Name	Type	Role
<i>Corpus construction</i>			
$F \in \{E, B, A\}$	field subset	category	restrict $\mathfrak{S}$ to Economics (E), Business (B), or all OAS (A); see Table 3
$t_x \in \{1, \dots, 7\}$	time window	integer	census and target year ranges; see Table 3
τ_U	institution threshold	integer	minimum mean annual works to retain an institution
<i>Matrix construction</i>			
$\rho \in \{0, 1\}$	reference normalisation	binary	0: a work contributes $\bar{R}$ units 1: a work contributes R_i units
$\mathbf{m} \in \{0, 1\}^4$	block mask	binary 4-vector	select active unit-type pairs (SS , SI , IS , II)
$\chi \in [0, 1]$	mixing weight	continuous	source–institution balance within each work
<i>Ranking</i>			
$\alpha \in [0, 1)$	damping factor	continuous	controls propagation through citation network ensures convergence

Spectral ranking. Let $r_p = \sum_q C_{pq}$ be the total outgoing citation flow from unit p , $\mathbf{D}_r = \text{diag}(r_p)$ the corresponding diagonal matrix. If $r_p = 0$ (a dangling unit with no outgoing citations) the corresponding row of $\mathbf{C}$ is zero and $\mathbf{H}$ is defined to have a zero row there; the boundary condition below then supplies the necessary probability mass. Let $\boldsymbol{\mu}$ be an initial preference, or boundary condition (Hubbell, 1965; Vigna, 2016), with $\mu_p = 1/N$ for all $p \in \mathfrak{P}$. The row-stochastic citation matrix is $\mathbf{H} = \mathbf{D}_r^{-1} \mathbf{C}$.

The spectral ranking $\boldsymbol{\pi}$ (see Vigna, 2016, for an overview) is defined as the solution to the damped eigenvector problem

$$\boldsymbol{\pi} = \alpha \mathbf{H}^\top \boldsymbol{\pi} + (1 - \alpha) \boldsymbol{\mu}, \quad \mathbf{1}^\top \boldsymbol{\pi} = 1, \quad \boldsymbol{\pi} \geq 0, \quad (5)$$

with damping factor $\alpha \in [0, 1)$. The prior $\boldsymbol{\mu}$ encodes the distribution over units weighted by scholarly output, and α controls the degree to which influence propagates through the citation network rather than

being absorbed by that prior. In the limit $\alpha \rightarrow 1$, equation (5) reduces to the principal eigenvector problem

$$\boldsymbol{\pi}^\infty = \mathbf{H}^\top \boldsymbol{\pi}^\infty, \quad \mathbf{1}^\top \boldsymbol{\pi}^\infty = 1, \quad \boldsymbol{\pi}^\infty \geq 0, \quad (6)$$

whose solution is unique when $\mathbf{H}$ is primitive (irreducible and aperiodic). Geller (1978) shows that $\boldsymbol{\pi}^\infty$ is then the stationary distribution of the Markov chain with transition matrix $\mathbf{H}$, and that it coincides with the *influence weight* of Pinski and Narin (1976) scaled by the reference count proportion $r_p / \sum_q r_q$.

The spectral ranking $\boldsymbol{\pi}$ reflects the total citation flow received by each unit and therefore depends on unit size. *Prestige per work* (Palacios-Huerta and Volij, 2004),

$$\mathbf{v}(\alpha) = \mathbf{A}^{-1} \boldsymbol{\pi}(\alpha), \quad \mathbf{A} = \text{diag}(a_p), \quad (7)$$

removes this dependence, where $a_p = |\{i : w_i \in s\}|$ for source $p = s$ and $a_p = \sum_i \omega_{iu}$ for institution $p = u$ via (1) are the work-counts, $\boldsymbol{\pi}(\alpha)$ denotes the solution to (5), and $\boldsymbol{\pi}^\infty$ the limiting case (6). Palacios-Huerta and Volij (2004) show that $\mathbf{v}$ is uniquely characterised by several reasonable axiomatic constraints (but see Serrano, 2004). Note that $\mathbf{A}^{-1}$ must be applied after convergence; applying it inside the iteration loop would iterate on $\mathbf{H}^\top \mathbf{A}$, which is not norm-preserving and yields a fixed point with no clean interpretation.

To compute $\boldsymbol{\pi}$ we define the modified matrix

$$\tilde{\mathbf{H}} = \alpha \mathbf{H} + (1 - \alpha) \boldsymbol{\mu} \mathbf{1}^\top, \quad (8)$$

so that the fixed-point condition $\boldsymbol{\pi} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}$ is equivalent to equation (5). Because $\boldsymbol{\mu} > 0$ and $0 < \alpha < 1$, every entry of $\tilde{\mathbf{H}}$ is strictly positive, ensuring it is a primitive row-stochastic matrix. By the Perron–Frobenius theorem, $\tilde{\mathbf{H}}^\top$ therefore has a unique stationary distribution $\boldsymbol{\pi}$, recovered by power iteration

$$\boldsymbol{\pi}^{(k+1)} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}^{(k)}, \quad (9)$$

converging to $\boldsymbol{\pi}$ from any strictly positive starting vector $\boldsymbol{\pi}^{(0)}$.

Bipartite structure. With $\mathbf{m} = (0, 1, 1, 0)$, equation (4) gives a block off-diagonal $\mathbf{H}$, forming a bipartite directed graph between sources and institutions. Since the product of row-stochastic matrices is row-stochastic, $\mathbf{H}^2$ is block-diagonal and row-stochastic, with one-mode projection blocks $\mathbf{M}_S = \mathbf{H}_{SI} \mathbf{H}_{IS}$ and $\mathbf{M}_I = \mathbf{H}_{IS} \mathbf{H}_{SI}$. The bipartite structure makes $\mathbf{H}$ periodic with period 2, so the undamped equation (6) need not have a unique solution; $\tilde{\mathbf{H}}$ in equation (8) restores primitivity for any $\alpha \in (0, 1)$. Partitioning equation (5) and eliminating $\boldsymbol{\pi}_I$ yields the Katz–Hubbell resolvent for sources,

$$(\mathbf{I} - \alpha^2 \mathbf{M}_S^\top) \boldsymbol{\pi}_S = (1 - \alpha)(\boldsymbol{\mu}_S + \alpha \mathbf{H}_{IS}^\top \boldsymbol{\mu}_I), \quad (10)$$

where the effective damping is α^2 and the effective prior transports institution mass back to sources through $\mathbf{H}_{IS}^\top$. The institution ranking is then recovered by a single matrix–vector product,

$$\boldsymbol{\pi}_I = \alpha \mathbf{H}_{SI}^\top \boldsymbol{\pi}_S + (1 - \alpha) \boldsymbol{\mu}_I. \quad (11)$$

$\mathbf{C}(\chi, \mathbf{m}, \rho)$ is a sparse matrix built from a weighted edge list $\mathcal{E}$ accumulated over $\mathfrak{R}$ with per-reference weight ρ_i , stored in CSR format, and row-normalised to $\mathbf{H}$. The pipeline $\mathfrak{R} \rightarrow \mathcal{E} \rightarrow \mathbf{C} \rightarrow \mathbf{H} \rightarrow \boldsymbol{\pi} \rightarrow \mathbf{v}$ is implemented using `scipy.sparse.csr_matrix` with parameter entry points detailed in the Supplement.

3 Corpus selection

Corpus extraction starts with a list of eligible sources. We require each source to have three features: an OpenAlex source identifier for corpus extraction, evidence from one of our journal registries that the source publishes economics or business research, and relatively high OpenAlex source topic density to exclude cross-disciplinary journals and other edge cases.

The journal registries are as follows. Harzing’s *71st Edition of the Journal Quality List* (Harzing, 2024, JQL) assembles peer assessments of journal quality across business and economics disciplines. Its scope is

confined to economics and business so no filtering is applied. The *ERA 2023 Submission Journal List* (Australian Research Council, 2022, SJL) assigns up to three Field of Research (FoR) codes to each entry. From the full list of 26,241 journals we retain those having all FoR codes falling within Division 35 (Commerce, Management, Tourism and Services) or Division 38 (Economics). This applies to two-digit ‘division’ codes and all four-digit ‘group’ codes (3501–3509, 3801–3803, 3899). After filtering, 1,602 journals remain. *Clarivate Web of Science Journal Citation Reports* (Clarivate Analytics, 2026, JCR) assign each journal a Web of Science category. We retain those in Economics, Management, Business, Transportation or Business, Finance. After filtering, 1,413 journals remain.

Every journal in JQL, SJL, and MJL has at least one ISSN, and may have several. These registry ISSNs are joined independently to the February 2026 OpenAlex sources snapshot in using its list of ISSNs as the matching key. Journal titles are examined to identify around 30 additional matches where the OpenAlex source does not have an ISSN. The list of OpenAlex sources that are matched is designated OAS*. The long list OAS* includes sources publishing works in disciplines beyond economics and business. We use OpenAlex source-topic assignments to exclude sources having a minority of works in economics/business. A source has work-topic counts for up to five topics, each within a single sub-field, field, and domain. We accumulate the work-topic counts for OpenAlex Field 14 (Economics, Econometrics and Finance) and Field 20 (Business and Management) and define the work-topic density as the ratio of this count to the total topic count in the source. Sources are dropped if this ratio is below 50%. The 1,659 surviving sources, labeled OAS, are used to extract the corpus. Matching outcomes are summarised in Table 2.

Table 2: Registry source matches to OpenAlex source identifiers

Register	Journals	Matched	Match rate	Unmatched
JQL (Harzing’s journal quality list)	842	839	99.6%	3
MJL (Clarivate master journal list)	1,413	1,392	98.5%	21
SJL (ERA submission journal list)	1,602	1,388	86.6%	214
Registries pairwise overlap				
MJL $\cap$ SJL				783 sources
MJL $\cap$ JQL				566 sources
SJL $\cap$ JQL				424 sources
MJL $\cap$ SJL $\cap$ JQL				393 sources
OpenAlex topic filtering				
MJL $\cap$ SJL $\cap$ JQL $\cap$ OA (OAS* long-list)				2,380 sources
OA sources with topics specifically in Economics or Commerce (OAS)				1,659 sources

Each corpus extraction is parameterised by a field subset F and a time window t_x . The field subset restricts the source set $\mathfrak{S}$ to sources whose OpenAlex topic counts fall exclusively in Field 14 ($F = E$), exclusively in Field 20 ($F = B$), or in either field ($F = A$, all OAS sources). The time window encodes the census and target year ranges via $t_x \in \{1, \dots, 7\}$; the seven cases are listed in Table 3. The five symmetric windows correspond to conventional five-year citation windows; cases 6 and 7 are asymmetric windows analogous to the Journal Impact Factor and reference spectroscopy respectively.

We extract an initial corpus of works in the OAS source list ($F = A$, $t_x = 5$), by retaining article and review types with publication years between 2020 and 2024 inclusive and excluding works that are paratext or retracted, or without an authorship entity. The institution retention curve Fig. 1 implies that the preferred minimum cut for works per institution per year could be chosen as $\tau_U = 10$, leaving approximately 1,900 institutions and 75% of the works. The authorships and references for the initial corpus are then attached and the value of ω_{iu} computed for each work. The characteristics of the initial data set are exhibited in Table 4.

4 Results

Baseline model. [Placeholder.]

Table 3: Corpus construction parameters: field subsets (F) and time windows (t_x).

Panel A : Field subsets (F)

F Sources included in $\mathfrak{S}$

E Topic counts exclusively in Field 14 (Economics, Econometrics and Finance)

B Topic counts exclusively in Field 20 (Business, Management and Accounting)

A All OAS sources; topic counts in Fields 14 and/or 20 (*baseline*)

Panel B : Time windows (t_x)

t_x	Census window $[t_0^c, t_1^c]$	Target window $[t_0^t, t_1^t]$	Label
1	2000–2004	2000–2004	Symmetric 5-year
2	2005–2009	2005–2009	Symmetric 5-year
3	2010–2014	2010–2014	Symmetric 5-year
4	2015–2019	2015–2019	Symmetric 5-year
5	2020–2024	2020–2024	Symmetric 5-year (<i>baseline</i>)
6	2024	2000–2024	JIF analogue
7	2020–2024	2020	Reference spectroscopy

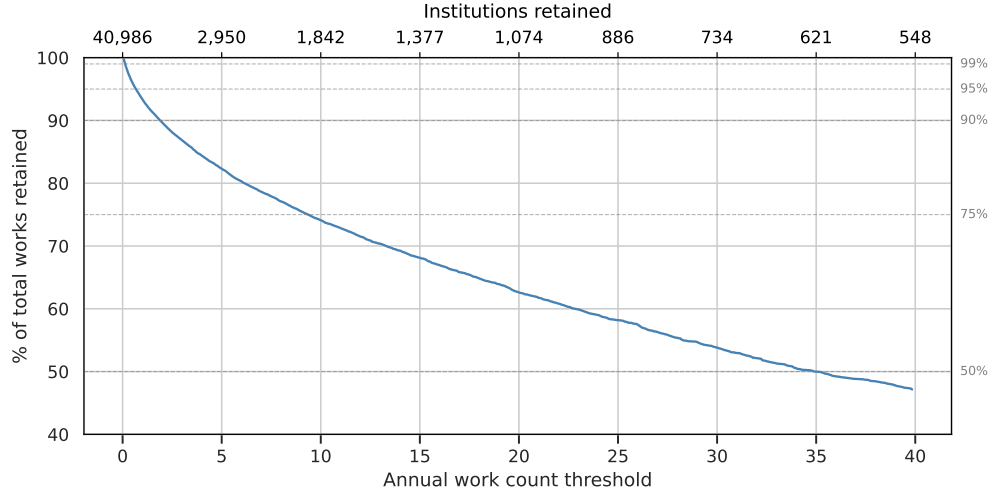


Figure 1: Institution retention curve

Table 4: Corpus features by year ($\tau_U > 10$). Upper panel: unique-item counts. Lower panel: D_1 and D_9 deciles of per-entity distributions. The 2000–24 column pools the full period.

Quantity	2000		2024		2000–24	
	count		count		count	
Works	16,190		59,802		966,687	
Sources	669		1,327		1,624	
Institutions	1,329		1,739		1,742	
Reference counts (out-degree)	6,304		1,546,508		14,897,377	
Citation counts (in-degree)	580,260		31,558		14,751,724	
Quantity	D_1	D_9	D_1	D_9	D_1	D_9
Works per source	3.0	49.0	5.0	84.4	32.0	1283.9
Institutions per work	1.0	3.0	1.0	4.0	1.0	4.0
References per work (out-degree)	1.0	3.0	5.0	60.0	2.0	44.0
Citations per work (in-degree)	2.0	92.0	1.0	4.0	1.0	44.0

Table 5: Parameters and values explored. Baseline values in italics.

Symbol	Name	Baseline	Values explored
<i>Corpus construction (see Tables 1 and 3)</i>			
t_x	Time window	<i>5</i>	1, 2, 3, 4, <i>5</i> , 6, 7
F	Field subset	<i>A</i>	<i>E</i> , <i>B</i> , <i>A</i>
τ_U	Institution threshold	<i>[TBD]</i>	[TBD]
<i>Matrix construction</i>			
ρ	Reference normalisation	<i>1</i>	<i>1</i> (full), 0 (fixed)
$\mathbf{m}$	Block mask	(1, 0, 0, 0)	(1, 0, 0, 0), (0, 1, 1, 0), (1, 1, 1, 1)
χ	Mixing weight	<i>0.50</i>	0.00, 0.25, <i>0.50</i> , 0.75, 1.00
<i>Ranking</i>			
α	Damping factor	<i>0.85</i>	0.50, 0.75, <i>0.85</i> , 0.95

Parameter sensitivity. [Placeholder: $\alpha \in \{0.50, 0.75, 0.85, 0.95\}$; $\chi \in \{0.00, 0.25, 0.50, 0.75, 1.00\}$; $\mathbf{m} \in \{(1, 0, 0, 0), (0, 1, 1, 0), (1, 1, 1, 1)\}$; $\rho \in \{0, 1\}$; τ sweep.]

Bootstraps. [Placeholder.]

5 Discussion

Face validity. [Placeholder.]

Content validity. [Placeholder.]

Criterion validity — concurrent and predictive. [Placeholder.]

Construct validity. [Placeholder.]

6 Prospects

[Placeholder.]

7 Addenda

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